STUDENT PROJECTS LABORATORY - LOCKOUT / TAGOUT [LOTO] PROCEDURE

ENERGY ISOLATION SPECIFICATION | ASSET: FAB-05 | TAG: BLAST-PRESS-10T20T

Abrasive Blast Cabinet & Hydraulic Shop Presses (Enerpac 10T & Redline 20T)

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Notification Channel: Max Wright - Teams / Text Message | Zone: Zone 5: Fabrication, Sheet Metal & Abrasive Finishing (FAB)

STEP 1 - NORMAL MACHINE SHUTDOWN PROCEDURE

Blast Cabinet: Release trigger, toggle light/dust collector switch OFF (Top), close wall air drop ball valve. Hydraulic Presses: Slowly open hydraulic release valve to retract ram fully.

STEP 2 - HAZARDOUS ENERGY SOURCES, MAGNITUDES & ISOLATION POINTS

Source #1: Electrical (Blast Cabinet) (120V AC)

Isolation Location: Vacuum / cabinet light power cord.

Lockout Procedure: Apply Rotating Electrical Plug Lockout over vacuum/light power cord; wait 60 seconds for dust collector impeller to stop; toggle switch ON to verify zero energy.

Source #2: Pneumatic (Blast Cabinet) (100 PSI Compressed Air)

Isolation Location: Orange wall air valve & cabinet regulator.

Lockout Procedure: Turn orange air valve OFF, disconnect airline from regulator, and depress foot pedal/trigger inside cabinet until regulator gauges read 0 PSI.

Source #3: Hydraulic / Mechanical (Enerpac 10T & Redline 20T Presses) (Up to 10,000 PSI Hydraulic / 20 Tons Force)

Isolation Location: Hydraulic hand/air pump release valve & H-frame bed support pins.

Lockout Procedure: Open hydraulic release valve to retract cylinder ram completely (0 PSI on GF10P gauge) and verify both H-frame bed support pins are fully seated.

STEP 3 - STORED / RESIDUAL ENERGY DISSIPATION

Blast Cabinet: Wait 60s for impeller stop + depress foot pedal to bleed air to 0 PSI.
Presses: Retract hydraulic ram until gauge reads 0 PSI.

STEP 4 - ZERO-ENERGY VERIFICATION ('TRY' STEP)

Toggle cabinet switch ON (no light/vacuum), confirm air gauge reads 0 PSI, and confirm hydraulic press GF10P gauge reads 0 PSI.